

EE3621: Digital Signal Processing

Lecture 5: Z-Transform, ROC & Common Pairs

(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 08:00 (8 mins):** Motivation: Z-transform as a generalization of the DTFT ($z = re^{j\omega}$) and convergence proof.
 - **08:00 – 18:00 (10 mins):** The Region of Convergence (ROC): Detailed properties, causal vs. anti-causal vs. two-sided signals.
 - **18:00 – 30:00 (12 mins):** Properties of the Z-Transform with full algebraic proofs.
 - **30:00 – 35:00 (5 mins):** Common Z-Transform Pairs and Pole-Zero plots.
 - **35:00 – 40:00 (5 mins):** Checkpoint and Q&A.
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2 Definition & Mathematical Motivation

The Z-transform of a sequence $x[n]$ is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad (1)$$

Expressing z in polar form $z = re^{j\omega}$:

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n} \quad (2)$$

This shows that the Z-transform corresponds to the DTFT of $x[n]$ scaled by r^{-n} . Evaluating $X(z)$ on the unit circle ($|z| = 1$) yields the DTFT $X(e^{j\omega})$. The Z-transform converges if the scaled sequence is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |x[n]r^{-n}| < \infty \quad (3)$$

The set of values of z satisfying this condition forms the Region of Convergence (ROC).

3 Properties of the Region of Convergence (ROC)

- **Shape:** The ROC consists of concentric rings in the z -plane centered at the origin.
- **Poles:** The ROC does not contain any poles.
- **Causal (Right-sided) signals:** ROC is the exterior of a circle: $|z| > r_{max}$.
- **Anti-causal (Left-sided) signals:** ROC is the interior of a circle: $|z| < r_{min}$.
- **Two-sided signals:** ROC is an annular ring: $r_{min} < |z| < r_{max}$.
- **Stability:** A system is stable if and only if the ROC contains the unit circle ($|z| = 1$). For causal systems, this requires all poles to lie inside the unit circle ($r_{max} < 1$).

See Figure 1 for illustrations of causal and anti-causal ROC regions.

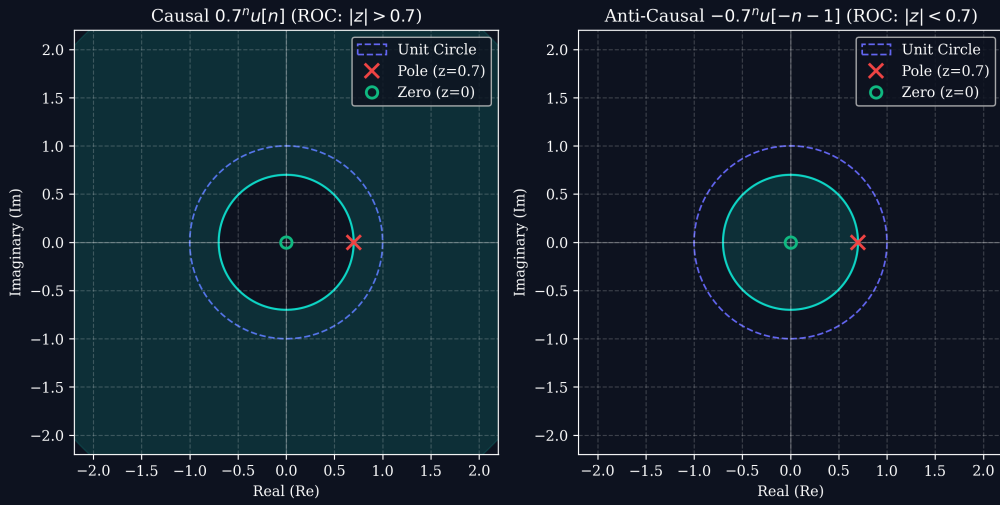


Figure 1: ROC shapes in the complex z -plane for causal and anti-causal exponential sequences.

4 Z-Transform Properties & Proofs

For $x[n] \leftrightarrow X(z)$ (ROC R_x) and $y[n] \leftrightarrow Y(z)$ (ROC R_y):

4.1 Time Shifting

$$x[n - n_0] \longleftrightarrow z^{-n_0} X(z) \quad (4)$$

Proof: Let $m = n - n_0$.

$$\sum_{n=-\infty}^{\infty} x[n - n_0] z^{-n} = \sum_{m=-\infty}^{\infty} x[m] z^{-(m+n_0)} = z^{-n_0} X(z) \quad (5)$$

4.2 Scaling in the Z-domain

$$a^n x[n] \longleftrightarrow X(a^{-1} z) \quad (6)$$

Proof:

$$\sum_{n=-\infty}^{\infty} a^n x[n] z^{-n} = \sum_{n=-\infty}^{\infty} x[n] (a^{-1} z)^{-n} = X(a^{-1} z) \quad (7)$$

4.3 Differentiation in the Z-domain

$$nx[n] \longleftrightarrow -z \frac{dX(z)}{dz} \quad (8)$$

Proof:

$$\frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} x[n](-n)z^{-n-1} = -z^{-1} \sum_{n=-\infty}^{\infty} nx[n]z^{-n} \implies -z \frac{dX(z)}{dz} = Z\{nx[n]\} \quad (9)$$

4.4 Initial Value Theorem

For causal $x[n]$:

$$x[0] = \lim_{z \rightarrow \infty} X(z) \quad (10)$$

Proof: $X(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots$ As $z \rightarrow \infty$, all terms with negative powers of z vanish.

5 Common Pairs and Annular ROC

Table 1 summarizes the primary Z-transform pairs. An annular ring representing a two-sided signal is illustrated in Figure 2.

Table 1: Common Z-transform pairs.

Signal $x[n]$	Z-Transform $X(z)$	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $

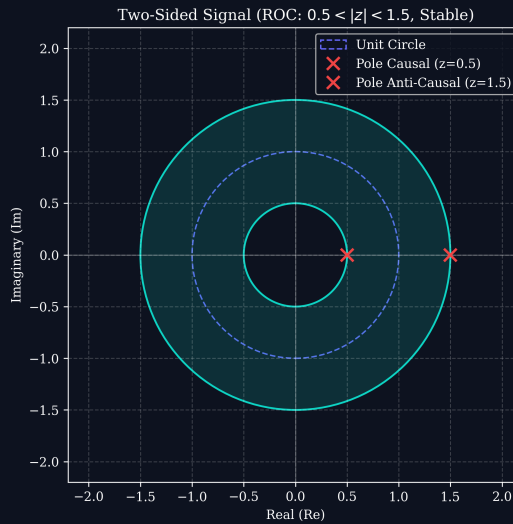


Figure 2: ROC of a stable two-sided sequence, forming an annular ring bounded by pole radii.